Note 7.4

A matrix which is both symmetric and skew-symmetric is a zero matrix.

Example 7.13

Express the matrix $A=\begin{bmatrix}1 & 3 & 5\\ -6 & 8 & 3\\ -4 & 6 & 5\end{bmatrix}$ as the sum of a symmetric and a skew-symmetric matrices.

Solution

$$A=\begin{bmatrix}1 & 3 & 5\\ -6 & 8 & 3\\ -4 & 6 & 5\end{bmatrix} \Rightarrow A^T=\begin{bmatrix}1 & -6 & -4\\ 3 & 8 & 6\\ 5 & 3 & 5\end{bmatrix}$$

$$\text{Let } P=\frac{1}{2}(A+A^T)=\frac{1}{2}\begin{bmatrix}2 & -3 & 1\\ -3 & 16 & 9\\ 1 & 9 & 10\end{bmatrix}$$

$$\text{Now } P^T=\frac{1}{2}\begin{bmatrix}2 & -3 & 1\\ -3 & 16 & 9\\ 1 & 9 & 10\end{bmatrix}=P$$

Thus, $P=\frac{1}{2}(A+A^T)$ is a symmetric matrix.

$$\text{Let } Q=\frac{1}{2}(A-A^T)$$

$$=\frac{1}{2}\begin{bmatrix}0 & 9 & 9\\ -9 & 0 & -3\\ -9 & 3 & 0\end{bmatrix}$$

$$\text{Then } Q^T=\frac{1}{2}\begin{bmatrix}0 & -9 & -9\\ 9 & 0 & 3\\ 9 & -3 & 0\end{bmatrix}=-Q$$

Thus $Q=\frac{1}{2}(A-A^T)$ is a skew-symmetric matrix.

$$A=P+Q=\frac{1}{2}\begin{bmatrix}2 & -3 & 1\\ -3 & 16 & 9\\ 1 & 9 & 10\end{bmatrix}+\frac{1}{2}\begin{bmatrix}0 & 9 & 9\\ -9 & 0 & -3\\ -9 & 3 & 0\end{bmatrix}$$

Thus A is expressed as the sum of symmetric and skew-symmetric matrices.

EXERCISE 7.1

(1) Construct an $m \times n$ matrix $A=[a_{ij}]$, where a_{ij} is given by

(i) $a_{ij}=\frac{(i-2j)^2}{2}$ with $m=2, n=3$ (ii) $a_{ij}=\frac{|3i-4j|}{4}$ with $m=3, n=4$

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